

Training Day 17 Report – Malware Evasion Techniques & Detection

Date: 04 July 2025

Module: Malware Obfuscation, Evasion & Signature-Based Detection

Mode: Theoretical + Tool-Based Lab

Main Focus of the Day

Day 18 explored **evasion techniques** used by malware to avoid detection and introduced **signature-based and behavior-based detection** strategies.

Topics Covered

- **Evasion Techniques:**
 - Code obfuscation
 - Packing and encryption
 - Anti-VM and anti-debugging techniques
 - Sleep/delay execution
- **Detection Mechanisms:**
 - Signature-based (AV engines, YARA rules)
 - Heuristics-based
 - Behavior-based (EDR systems)
- **Tools Introduced:**
 - YARA – Rule-based malware detection
 - UPX – Pack/unpack executables
 - IDA Free / Ghidra – Malware disassembly/decompilation
 - AV Sandboxing Logs (Defender, Kaspersky, etc.)

Practical Exercises

- Identified packed files using `upx -t sample.exe`
- Wrote and tested basic YARA rule to match malware pattern
- Observed how changing byte patterns evaded detection
- Simulated behavior-based alerts using Windows Defender logs

Key Takeaways

- Modern malware uses **sophisticated obfuscation** to evade static tools
- **YARA rules** provide a flexible detection framework for analysts

- **Unpacking and behavioral inspection** are critical in real-world cases
- Malware often hides in memory or uses polymorphism to stay undetected